# University of Ottawa

Department of Mathematics and Statistics

MAT 1302A : Mathematical Methods II

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First Midterm Exam – Version A

February 6, 2015

Surname _____ First Name _____

Student # _____ DGD _____

**Instructions:**

(a) You have 80 minutes to complete this exam.
(b) The number of points available for each question is indicated in square brackets.
(c) All work to be considered for grading should be written in the space provided. The reverse side of pages is for scrap work. If you find that you need extra space in order to answer a particular question, you should continue on the reverse side of the page and indicate this **clearly**. Otherwise, the work written on the reverse side of pages will not be considered for marks.
(d) Write your student number at the top of each page in the space provided.
(e) No notes, books, scrap paper, calculators or other electronic devices are allowed.
(f) You are strongly recommended to write in **pen**, not pencil.
(g) You may use the last page of the exam as scrap paper.

Good luck!

Please do not write in the table below.

| Question | 1 | 2 | 3 | 4 | 5 | 6 | Total |
|---|---|---|---|---|---|---|---|
| Maximum | 4 | 4 | 3 | 5 | 3 | 3 | 22 |
| Grade | | | | | | | |

1. [**4 points**] Determine if the following linear system is consistent or inconsistent. (If the system is consistent, you do not need to find the general solution.)

$$\begin{cases} x_2 + 4x_4 + x_5 &= 5 + 2x_1 + x_3 \\ x_2 + 2x_3 + 3x_4 + 3x_5 &= 2x_1 + 6 \\ 2x_1 - 5x_3 &= 2x_4 + 5x_5 - 7 + x_2 \end{cases}$$

**Solution:** The linear system in standard form is:

$$\begin{cases} -2x_1 + x_2 - x_3 + 4x_4 + x_5 = 5 \\ -2x_1 + x_2 + 2x_3 + 3x_4 + 3x_5 = 6 \\ 2x_1 - x_2 - 5x_3 - 2x_4 - 5x_5 = -7 \end{cases}$$

We row-reduce the augmented matrix to an echelon form:

$$\left[\begin{array}{ccccc|c} -2 & 1 & -1 & 4 & 1 & 5 \\ -2 & 1 & 2 & 3 & 3 & 6 \\ 2 & -1 & -5 & -2 & -5 & -7 \end{array}\right] \xrightarrow{R_2 \to R_2 - R_1} \left[\begin{array}{ccccc|c} -2 & 1 & -1 & 4 & 1 & 5 \\ 0 & 0 & 3 & -1 & 2 & 1 \\ 2 & -1 & -5 & -2 & -5 & -7 \end{array}\right]$$

$$\xrightarrow{R_3 \to R_3 + R_1} \left[\begin{array}{ccccc|c} -2 & 1 & -1 & 4 & 1 & 5 \\ 0 & 0 & 3 & -1 & 2 & 1 \\ 0 & 0 & -6 & 2 & -4 & -2 \end{array}\right]$$

$$\xrightarrow{R_3 \to R_3 + 2R_2} \left[\begin{array}{ccccc|c} -2 & 1 & -1 & 4 & 1 & 5 \\ 0 & 0 & 3 & -1 & 2 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

Since the rightmost column is not a pivot column, the linear system is consistent.

2. (a) **[3 points]** Find the general solution of the following vector equation. Write the answer in vector parametric form.

$$x_1\begin{bmatrix}-3\\3\\1\end{bmatrix}+x_2\begin{bmatrix}6\\-6\\-2\end{bmatrix}+x_3\begin{bmatrix}5\\-6\\-\frac{5}{3}\end{bmatrix}=\begin{bmatrix}0\\0\\0\end{bmatrix}$$

(b) **[1 point]** The values

$$x_1=25,\ x_2=5,\ \text{and } x_3=-30$$

form a solution for the linear system

$$\begin{bmatrix}-3&6&5\\3&-6&-6\\1&-2&-\frac{5}{3}\end{bmatrix}\begin{bmatrix}x_1\\x_2\\x_3\end{bmatrix}=\begin{bmatrix}-195\\225\\65\end{bmatrix}$$

*Without using row reduction*, find the general solution for the above linear system.Write your answer in vector parametric form.

**Hint.** Compare the coefficient matrix of the linear system in part (b) with the one in part (a).

**Solution:** (a) We row-reduce the augmented matrix.

$$\begin{bmatrix}-3&6&5&|&0\\3&-6&-6&|&0\\1&-2&-\frac{5}{3}&|&0\end{bmatrix}\xrightarrow[R_3\to R_3+\frac{1}{3}R_1]{R_2\to R_2+R_1}\begin{bmatrix}-3&6&5&|&0\\0&0&-1&|&0\\0&0&0&|&0\end{bmatrix}$$

$$\xrightarrow{R_2\mapsto -R_2}\begin{bmatrix}-3&6&5&|&0\\0&0&1&|&0\\0&0&0&|&0\end{bmatrix}\xrightarrow{R_1\to R_1-5R_2}\begin{bmatrix}-3&6&0&|&0\\0&0&1&|&0\\0&0&0&|&0\end{bmatrix}$$

$$\xrightarrow{R_1\to -\frac{1}{3}R_1}\begin{bmatrix}1&-2&0&|&0\\0&0&1&|&0\\0&0&0&|&0\end{bmatrix}$$

The linear system corresponding to the REF is

$$\begin{cases}x_1-2x_2=0\\x_3=0\end{cases}\Rightarrow\begin{cases}x_1=2x_2\\x_2=\text{free}\\x_3=0\end{cases}$$

Therefore the vector parametric form of the solution is

$$\begin{bmatrix}x_1\\x_2\\x_3\end{bmatrix}=x_2\begin{bmatrix}2\\1\\0\end{bmatrix}$$

(b) From the relation between the solution of a non-homogeneous system with its homogeneous form, and result of part (a), we obtain

$$\begin{bmatrix}x_1\\x_2\\x_3\end{bmatrix}=\begin{bmatrix}25\\5\\-30\end{bmatrix}+x_2\begin{bmatrix}2\\1\\0\end{bmatrix}$$

3. Calculate the following.

(a) **[1 point]** $3\begin{bmatrix}-1\\3\\2\end{bmatrix} - 2\begin{bmatrix}2\\0\\4\end{bmatrix}$

**Solution:** $\begin{bmatrix}-7\\9\\-2\end{bmatrix}$

(b) **[2 points]** $\begin{bmatrix}-2 & -4 & 2\\1 & 0 & -3\\-1 & 1 & 1\end{bmatrix}\begin{bmatrix}-1\\-2\\1\end{bmatrix}$

**Solution:** $\begin{bmatrix}12\\-4\\0\end{bmatrix}$

4. (a) **[3 points]** Determine the value of the parameter $t$ such that the following linear system has infinitely many solutions.

$$\begin{cases} x_1 & -x_2 & -3x_3 & = & 1 \\ 2x_1 & -x_2 & -3x_3 & = & -3 \\ x_1 & -x_2 & +tx_3 & = & 1 \end{cases}$$

**Solution:** We row-reduce the augmented matrix to an echelon form:

$$\left[\begin{array}{ccc|c} 1 & -1 & -3 & 1 \\ 2 & -1 & -3 & -3 \\ 1 & -1 & t & 1 \end{array}\right] \xrightarrow[R_3 \to R_3 - R_1]{R_2 \to R_2 - 2R_1} \left[\begin{array}{ccc|c} 1 & -1 & -3 & 1 \\ 0 & 1 & 3 & -5 \\ 0 & 0 & t+3 & 0 \end{array}\right]$$

We need the system to be consistent and have at least one free variable. Both of these conditions are satisfied if and only if $t = -3$.

(b) **[2 points]** Write down the general solution of the linear system for the value of $t$ that you find in part (a).

**Solution:** For $t = -3$ we row-reduce the augmented matrix to the reduced echelon form.

$$\left[\begin{array}{ccc|c} 1 & -1 & -3 & 1 \\ 0 & 1 & 3 & -5 \\ 0 & 0 & 0 & 0 \end{array}\right] \xrightarrow{R_1 \to R_1 + R_2} \left[\begin{array}{ccc|c} 1 & 0 & 0 & -4 \\ 0 & 1 & 3 & -5 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

The linear system corresponding to the REF is

$$\begin{cases} x_1 & & & = & -4 \\ & x_2 & +3x_3 & = & -5 \end{cases} \Rightarrow \begin{cases} x_1 & = -4 \\ x_2 & = -3x_3 - 5 \\ x_3 & = \text{free} \end{cases}$$

5. [**3 points**] Consider the vectors

$$\mathbf{u} = \begin{bmatrix} 1 \\ 2 \\ 2 \\ -\frac{1}{2} \end{bmatrix}, \ \mathbf{v}_1 = \begin{bmatrix} 2 \\ -2 \\ 4 \\ -1 \end{bmatrix}, \ \mathbf{v}_2 = \begin{bmatrix} 1 \\ -4 \\ 2 \\ -\frac{1}{2} \end{bmatrix}, \ \mathbf{v}_3 = \begin{bmatrix} -2 \\ 7 \\ -4 \\ 2 \end{bmatrix}$$

Determine if $\mathbf{u}$ belongs to $\text{Span}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$.

**Solution:** To answer this question we should row-reduce the following matrix:

$$\begin{bmatrix} 2 & 1 & -2 & 1 \\ -2 & -4 & 7 & 2 \\ 4 & 2 & -4 & 2 \\ -1 & -\frac{1}{2} & 2 & -\frac{1}{2} \end{bmatrix} \xrightarrow[R_4 \to R_4 + \frac{1}{2}R_1]{\substack{R_2 \to R_2 + R_1 \\ R_3 \to R_3 - 2R_2}} \begin{bmatrix} 2 & 1 & -2 & 1 \\ 0 & -3 & 5 & 3 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \xrightarrow{R_3 \rightsquigarrow R_4} \begin{bmatrix} 2 & 1 & -2 & 1 \\ 0 & -3 & 5 & 3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Since the rightmost column is not a pivot column, it follows that $\mathbf{u} \in \text{Span}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$.

6. **[3 points]** For each of the following statements, indicate if it is true or false. You will receive 1 point for every correct answer and lose .5 points for every incorrect answer (but you cannot receive a negative mark on this question).

(a) A consistent linear system with 5 equations and 5 variables always has a unique solution.

(b) If the span of two vectors $\mathbf{v}, \mathbf{w} \in \mathbb{R}^3$ is a line, then there exists a scalar $c \in \mathbb{R}$ such that $\mathbf{v} = c\mathbf{w}$ or $\mathbf{w} = c\mathbf{v}$.

(c) If $A$ is a $4 \times 5$ matrix and $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix}$ then the homogeneous linear system $A\mathbf{x} = \mathbf{0}$ has infinitely many solutions.

**Solution:** (a) False
(b) True
(c) True